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JAVA PROJECT REPORT ON

VEHICLE PARKING FEE COLLECTIOM

AND

REPORT GENERATION SYSTEM

Submitted by

VARNIKHA VANI N

25BCS260

Under the Guidance of

DHARANEESH

Technical Trainer

Technical Department, KG College of Arts and Science

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Semester: **I**

DECLARATION

I hereby declare that the project entitled "**PROJECT TITLE"** submitted in partial fulfillment of the requirements for the course **[Course Name]** is a **record of my original work** carried out under the guidance of **[Trainer Name]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Varnikha vani N

CERTIFICATE

This is to certify that the project entitled **"PROJECT TITLE"** is a **bona fide work** carried out by **[YOUR NAME] [Class/Batch]**, [**Department Name]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:10/04/2026

Submitted for the Viva-Voce Examination held on : 10.04.2026

Guide: **Head of the Department**(Technical Trainer, KGCAS) (Technical Head, KGCAS)

DHARANEESH BOOPALAN

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ABSTRACT

The Vehicle Parking Fee Collection and Report Generation System is a Java-based application developed to automate the process of managing parking operations and fee collection. In many parking areas, fee collection is done manually, which may lead to errors, delays, and difficulty in maintaining records. This system provides a computerized solution to efficiently handle parking entries, exits, and fee calculations.

The system allows the user to enter vehicle details such as vehicle number, type, entry time, and exit time. Based on the duration of parking, the system automatically calculates the parking fee. It also generates reports that include daily collections, vehicle details, and total revenue.

The application is developed using Core Java concepts such as classes, objects, loops, and conditional statements. The system improves accuracy, reduces manual work, and provides a faster and more efficient

parking management solution.

1. INTRODUCTION

.In today's fast-growing urban environment, the number of vehicles on roads has increased significantly. This rapid growth has created a high demand for efficient parking management systems in places such as shopping malls, hospitals, offices, airports, and public parking areas. Managing parking manually in such environments is often difficult, time-consuming, and prone to human errors.

Traditional parking systems rely on manual entry of vehicle details and handwritten fee collection. This approach can lead to several issues such as incorrect fee calculation, loss of records, traffic congestion at entry and exit points, and inefficient utilization of parking space. Additionally, maintaining daily records and generating reports manually becomes a complex task, especially when dealing with a large number of vehicles.

To overcome these challenges, the Vehicle Parking Fee Collection and Report Generation System is developed as an automated solution. This system uses computer-based technology to manage parking operations efficiently. It records vehicle details at the time of entry, tracks the duration of parking, calculates the parking fee automatically, and generates accurate reports.

The system ensures that all calculations are performed correctly based on predefined rates, reducing the chances of human error. It also helps in maintaining digital records of vehicles, which can be easily accessed for future reference. The report generation feature provides useful information such as total number of vehicles parked, total revenue collected, and parking usage statistics.

This project is implemented using Java programming language, which is widely used for developing robust and scalable applications. The system makes use of core Java concepts such as classes, objects, loops, and conditional statements to perform its operations efficiently.

Overall, this system improves the efficiency, accuracy, and reliability of parking management. It reduces manual workload, saves time, and provides a better experience for both parking operators and users. The project also serves as a practical example of applying programming concepts to solve real-world problems.

1.1 Objective

.The main objective of the Vehicle Parking Fee Collection and Report Generation System is to develop an efficient and automated solution for managing parking operations. The system is designed to simplify the process of vehicle entry, exit, fee calculation, and report generation using Java programming.

The specific objectives of this project are:

To design and develop a computerized parking management system

To record vehicle details such as vehicle number, entry time, and exit time

To calculate parking fees automatically based on the duration of parking

To reduce manual errors in fee calculation and record maintenance

To minimize time consumption during vehicle entry and exit

To provide a user-friendly interface for easy operation

To maintain accurate and organized records of parked vehicles

To generate reports showing total vehicles and total fee collected

To improve efficiency in managing parking spaces

To demonstrate the use of Core Java concepts such as classes, objects, loops, and conditional statements

To provide a basic foundation for developing advanced parking systems in the future

1.2 Overview

.The Vehicle Parking Fee Collection System is a menu-driven application. It allows users to:

Enter vehicle details at the time of entry

Record exit time of vehicles

Automatically calculate parking fees

Generate reports of parked vehicles and total collection

The system can be implemented as a console-based program or GUI-based application.

1.3 Features

.The Vehicle Parking Fee Collection and Report Generation System includes the following main features:

• Vehicle Entry Recording – Records vehicle number and entry time when a vehicle enters the parking area.

• Vehicle Exit Tracking – Captures exit time of the vehicle and determines the total parking duration.

• Automatic Fee Calculation – Automatically calculates the parking fee based on the number of hours parked.

• Time-Based Billing – Calculates charges according to parking duration (e.g., ₹20 per hour).

• Report Generation – Generates reports showing vehicle details and total parking fee collection.

• Data Accuracy – Reduces human errors by performing automatic calculations and maintaining correct records.

• User-Friendly Interface – Provides a simple and easy-to-use system for entering and viewing parking details.

• Fast Processing – Quickly processes vehicle data and displays parking fee without delay.

• Record Maintenance – Stores parking details for future reference and tracking.

1. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

.The Vehicle Parking Fee Collection and Report Generation System includes the following main features:

• Vehicle Entry Recording – Records vehicle number and entry time when a vehicle enters the parking area.

• Vehicle Exit Tracking – Captures exit time of the vehicle and determines the total parking duration.

• Automatic Fee Calculation – Automatically calculates the parking fee based on the number of hours parked.

• Time-Based Billing – Calculates charges according to parking duration (e.g., ₹20 per hour).

• Report Generation – Generates reports showing vehicle details and total parking fee collection.

• Data Accuracy – Reduces human errors by performing automatic calculations and maintaining correct records.

• User-Friendly Interface – Provides a simple and easy-to-use system for entering and viewing parking details.

• Fast Processing – Quickly processes vehicle data and displays parking fee without delay.

• Record Maintenance – Stores parking details for future reference and tracking.

2.2 Project Overview

.The Vehicle Parking Fee Collection and Report Generation System is a Java-based application designed to automate parking management.

The system allows users to:

• Enter vehicle details such as vehicle number and entry time

• Record exit time of vehicles

• Calculate parking duration automatically

• Compute parking fee based on time

• Generate reports of vehicles and total collection

The system simplifies parking operations by reducing manual work and improving accuracy. It ensures faster processing and better record management.

This project is developed using Core Java concepts and can be implemented as a console-based or GUI-based application.

2.3 Scopes and Limitations

Scope

• Manages vehicle entry and exit

• Calculates parking fees automatically

• Generates reports of vehicles and total collection

• Useful for small parking areas like malls and offices

Limitations

• No database storage

• No online payment system

• Limited to small-scale use

• Requires manual time entry

1. SYSTEM DESIGN

.System design is the process of planning the structure and working of the system before implementation. It divides the system into different modules to make the program simple and efficient. The Vehicle Parking System is designed as a menu-driven application for easy use.

Modules

The system is divided into the following modules:

• Vehicle Entry Module

• Order Selection Module

• Bill Calculation Module

• Invoice Generation Module

• Exit Module

1. Order Selection Module

This module allows the user to enter vehicle details.

• Accepts vehicle number

• Accepts entry and exit time

• Stores input data temporarily

2. Bill Calculation Module

This module calculates the parking fee.

• Calculates parking duration (Exit Time – Entry Time)

• Calculates fee based on time (₹20 per hour)

• Ensures accurate billing

3. Invoice Generation Module

This module displays the final parking details.

• Shows vehicle number

• Shows parking duration

• Shows total parking fee

• Displays formatted output (bill)

4. Exit Module

This module allows the user to close the system.

• Ends the program safely

• Stops further operations

1. METHODOLOGY

Methodology explains the process followed to develop the Vehicle Parking Fee Collection and Report Generation System. A step-by-step approach is used to make the system simple, accurate, and efficient.

The development process includes the following stages:

• Problem Analysis

• System Design

• Program Development

• Testing and Validation

• Result and Documentation

4.1. Problem Analysis

In this phase, the problems in the existing manual parking system are studied. Manual systems require more time and may lead to errors in fee calculation and record maintenance.

The main problems identified are:

• Incorrect parking fee calculation

• Delay in vehicle entry and exit process

• Difficulty in maintaining parking records

• More manual work and less efficiency

Based on these problems, the need for an automated parking system was identified.

4.2. System Design

In this stage, the overall structure of the system is planned. The system is divided into different modules so that each module performs a specific function.

The modules include:

• Vehicle Entry Module

• Order Selection Module

• Bill Calculation Module

• Invoice Generation Module

• Exit Module

This design helps to make the system easy to understand and implement.

4.3. Program Development

After designing the system, the program is developed using Java programming language. Core Java concepts are used to build the application.

The following concepts are used:

• Classes and Objects

• Loops

• Conditional Statements

• User Input Handling

• Output Display

The program accepts vehicle details, calculates the parking duration, computes the fee, and displays the final result.

4.4. Testing and Validation

Testing is done to ensure that the system works correctly and produces accurate results. Different test cases are used to check the performance of the system.

Example Test Cases:

Test Case	Expected Result
Enter vehicle number	vehicle number accepted correctly
Enter entry time	Entry time recorded correctly
Enter exit time	exit time recorded correctly
Calculate parking fee	Correct fee displayed

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

The system was successfully developed and tested. It records vehicle details, calculates parking duration, and displays the correct parking fee.

The program works efficiently and reduces manual effort. It also handles invalid inputs properly.

The documentation includes source code, input and output, testing details, and system explanation.

4.6. Algorithm

Algorithm for the Vehicle parking Management:

1. Start
2. Enter vehicle details
3. Record entry time
4. Enter exit time
5. Calculate parking duration
6. Calculate fee
7. Display bill
8. Store data
9. Generate report
10. End

4.7. Flowchart

Start

↓

Enter Vehicle Number

↓

Enter Entry Time

↓

Enter Exit Time

↓

Check if Exit Time < Entry Time

→ Yes → Display “Invalid Time” → End

→ No ↓

Calculate Parking Duration

↓

Calculate Parking Fee

↓

Display Parking Details

↓

End

1. SOURCE CODE

```java
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("=== Vehicle Parking System ===");

        System.out.print("Enter Vehicle Number: ");
        String vehicleNo = sc.next();

        System.out.print("Enter Entry Time (hour): ");
        int entry = sc.nextInt();

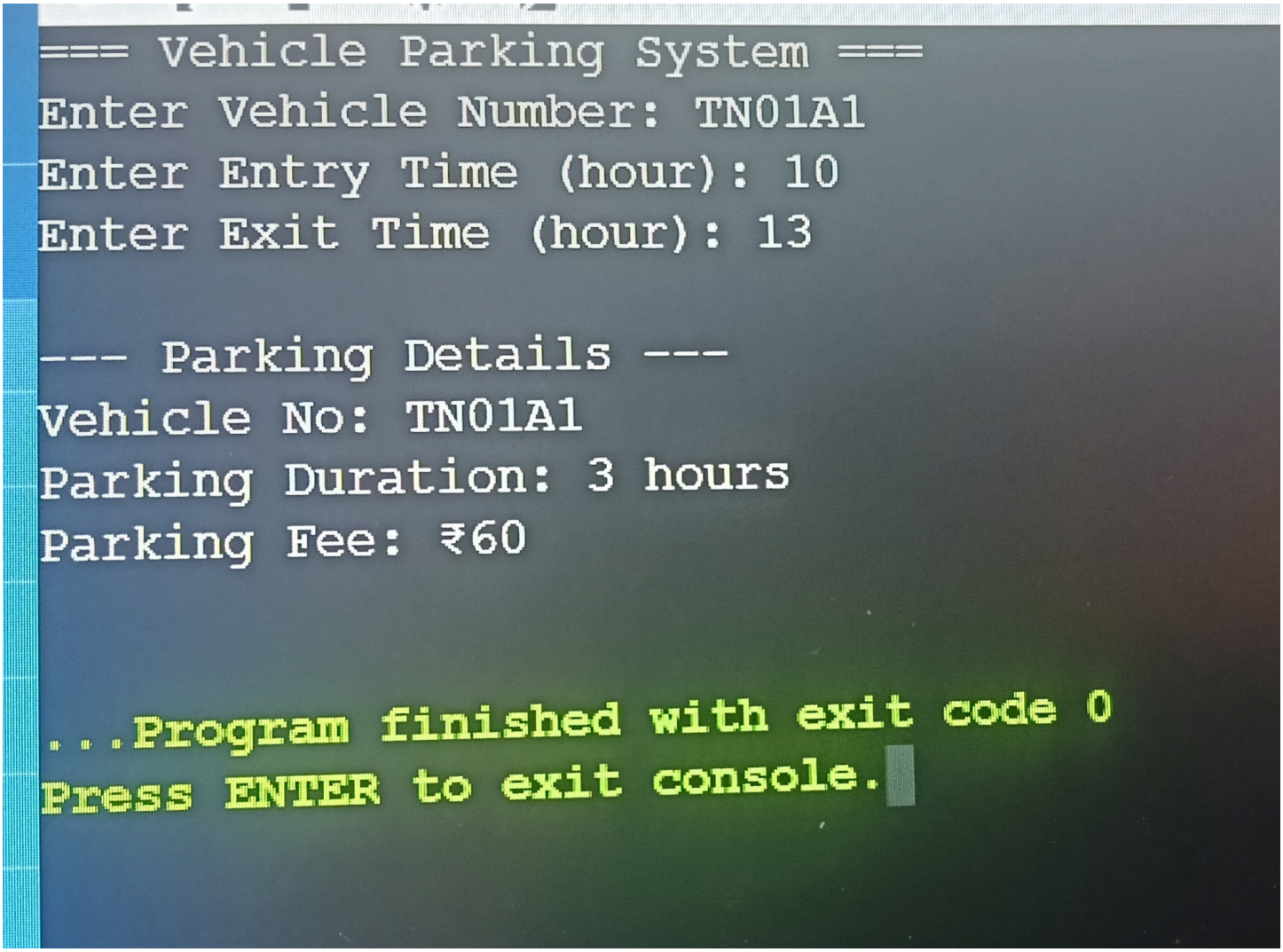
        System.out.print("Enter Exit Time (hour): ");
        int exit = sc.nextInt();

        int hours = exit - entry;
        if (hours < 0) {
            System.out.println("Invalid Time!");
        } else {
            int fee = hours * 20; // ₹20 per hour

            System.out.println("\n--- Parking Details ---");
            System.out.println("Vehicle No: " + vehicleNo);
            System.out.println("Parking Duration: " + hours + " hours");
            System.out.println("Parking Fee: ₹" + fee);
        }

        sc.close();
    }
}
```

INPUT AND OUTPUT SCREENS



DISCUSSION & CONCLUSION

7.1 Discussion

The Vehicle Parking Fee Collection and Report Generation System was developed to automate parking operations. The system records vehicle details, calculates parking duration, and generates the parking fee accurately.

The program was tested with different inputs and produced correct results. It reduces manual work and improves efficiency in parking management.

7.2 Conclusion

The system successfully automates the parking process and ensures accurate fee calculation. It is simple, fast, and easy to use.

This project demonstrates the practical use of Java programming in real-life applications. It can be further improved by adding features like database storage and online payment.

REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – Java: The Complete Reference

Websites

• https://www.oracle.com/java

• https://www.geeksforgeeks.org

• https://www.javatpoint.com

• https://www.w3schools.com